

Making Abstract Aesthetics Experiential: An Embodied Cultural Movement Framework for Intangible Cultural Heritage Interaction

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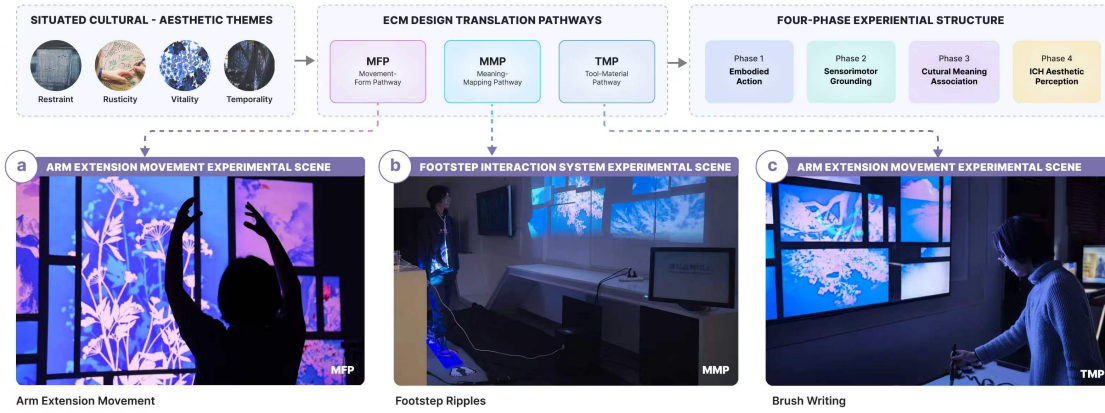


Fig. 1. Overview of the embodied interaction experience in the Blue Calico Realm interactive installation. Three ECMF-guided interactive scenes demonstrate how contextualized ICH aesthetic themes can be explored through movement-based and tool-mediated interaction.

A core challenge in intangible cultural heritage (ICH) digitization is translating abstract aesthetics—such as restraint and temporality—into embodied interactive experiences. Existing research lacks systematic embodied design approaches for abstract aesthetics that cannot be readily represented through explicit movement forms. We propose the Embodied Cultural Movement Framework (ECMF), which provides three design pathways for translating abstract aesthetics—movement form, meaning mapping, and tool-material relationships—and organizes experience through four phases. Guided by ECMF, we developed the Blue Calico Realm interactive installation and conducted a within-subjects evaluation ($N = 22$). Participants in the embodied condition reported significantly higher presence and scores across four ICH aesthetic perception dimensions compared with low-embodiment condition. Qualitative findings further revealed cultural associations, connections between bodily perception and prior experience, and culturally familiar patterns of tool use. ECMF provides a structured design approach for exploring how abstract ICH aesthetics can be translated into embodied interaction.

CCS CONCEPTS • Human-centered computing → HCI design and evaluation methods • Human-centered computing → Empirical studies in HCI • Applied computing → Arts and humanities

Additional Keywords and Phrases: Intangible Cultural Heritage, Embodied Cognition, Abstract Aesthetic Perception, Embodied Cultural Movement

1 INTRODUCTION

Digital technologies have enabled new forms of engagement with intangible cultural heritage (ICH), particularly for practices with recognizable movements, rituals, and production processes. For ICH practices with clearly identifiable embodied forms—such as ethnic dance, traditional rituals, and oral performance—existing research has extensively expanded digital experiences through multisensory immersion and interactive narratives [18,30], enabling users to engage with movement rhythms, production processes, and cultural contexts. Recent embodied interaction systems—including

mixed-reality experiences grounded in Chan meditation movements [12,13] and immersive VR environments for craft-making practice [18]—demonstrate that movement forms inherent to these cultural practices can effectively support users' engagement with ICH content.

However, when the design focus shifts toward the abstract aesthetics embedded in traditional crafts—such as rusticity, restraint, and temporality—these qualities cannot be directly derived from existing movement forms, cannot be fully captured through visual reconstruction alone [8,19,31,33] or by reproducing movement patterns from cultural practices [12,13,18]. The aesthetic meaning of traditional crafts is not only manifested in final patterns and artifact forms, but is closely tied to material transformations, tool manipulation, and bodily action [15]. Research has shown that bodily action and sensorimotor experience can provide an experiential foundation for understanding cultural content and abstract meaning that extends beyond pure representation [7,3,16,20]. Therefore, systematically translating abstract ICH aesthetics into bodily perceivable interactive forms constitutes the core design problem addressed in this paper.

Building on the view that bodily experience can contribute to meaning construction beyond visual representation, this study draws on embodied cognition theory and Peircean triadic semiotics to propose the Embodied Cultural Movement Framework (ECMF) to organize the design translation of abstract aesthetics into embodied interaction. Rather than reducing bodily movement to a simple content-triggering input, the framework conceptualizes the body as a medium for connecting action with cultural meaning. It provides three design pathways—movement form, meaning mapping, and tool–material relationships—and organizes the experiential relationship between bodily action and aesthetic perception through four phases: Embodied Action, Sensorimotor Grounding, Cultural Meaning Association, and ICH Aesthetic Perception.

Taking the Blue Calico Realm immersive interactive installation as a case study, this study translates four contextualized aesthetic themes—Restraint, Rusticity, Vitality, and Temporality—extracted from blue calico cultural materials into embodied interactions involving bodily action and tool use. It then evaluates the experiential differences between ECMF-guided embodied interaction and low-embodiment interaction through a within-subjects comparative experiment (EC vs. LEC, $n = 22$). This study addresses the following core research question: Compared with low-embodiment interaction that maintains the same primary audiovisual content, how do users' perceptions of abstract ICH aesthetics differ under ECMF-guided embodied interaction?

The specific contributions include:

- (1) This study contributes the Embodied Cultural Movement Framework (ECMF), a design-oriented framework that organizes the translation of abstract ICH aesthetics into embodied interaction through three complementary design pathways, while conceptualizing related experiential processes through a four-phase experiential structure.
- (2) Guided by the proposed ECMF, we develop Blue Calico Realm, an embodied interaction installation that operationalizes the translation from cultural aesthetic themes to bodily actions and multimodal feedback.
- (3) We provide empirical evidence on experiential differences associated with ECMF-guided embodied interaction through quantitative and qualitative evaluation, and derive design implications for ICH interaction design.

2 RELATED WORK

2.1 Embodied Cognition and Abstract Meaning Comprehension

Embodied cognition theory posits that cognitive processes are not built entirely upon disembodied, abstract symbols, but are fundamentally grounded in continuous sensorimotor interactions between the body and its physical environment [10]. Interactions between the body and material surroundings actively shape cognitive processing, while the manipulation of specific tools gradually incorporates them into body schemas, thereby reshaping individual's modes of cognition [21].

At the cognition level, theories such as Barsalou et al.'s Perceptual Symbol Systems (PSS) theory and Lakoff et al.'s Conceptual Metaphor Theory further explain how sensorimotor experiences contribute to abstract meaning construction [2,3,24]. PSS theory suggests that conceptual representations are grounded in modality-specific sensorimotor memory traces rather than purely amodal symbols. When bodily action reactivates these traces, it triggers mental simulation of situated concepts, thereby grounding abstract semantic understanding in experiential content. Similarly, Conceptual Metaphor Theory proposes that concrete sensorimotor schemas (image schemas) can be mapped onto abstract cognitive domains, providing an experiential basis for understanding abstract meaning. Accordingly, the understanding of cultural meaning involves perceptual experience, emotional response, and pre-existing cultural schemas.

Empirical research in digital heritage has also reported findings consistent with these theoretical perspectives. Bertrand et al. [5] integrated literary narrative with embodied presence in mixed reality cultural heritage experiences, in which participants shifted from passive observation toward more active meaning construction. This suggests that bodily action may provide an experiential basis for understanding cultural content that differs from purely visual observation.

Nevertheless, while embodied cognition elucidates how bodily experience participates in conceptual understanding, Peircean triadic semiotics provides a complementary lens for examining how bodily experience connects with specific cultural objects and meaning [32]. Within Peirce's semiotic framework, the sign process comprises the sign, object, and interpretant, and can manifest in relational modes such as iconicity, indexicality, and symbolicity. When applied to embodied interaction design, bodily actions can establish iconic connections through similarities in movement form, indexical connections through causal or contiguous relations between action and feedback, or symbolic connections through existing cultural conventions and shared meanings.

2.2 Intangible Cultural Heritage and Embodied Interaction

In the context of cultural heritage, Hu et al. [17] proposed a design paradigm grounded in embodied cognition, conceptualizing cultural heritage experiences through the interrelationship among heritage environment, body constitution, and cognitive activities. However, how such an embodied cognition perspective can be operationalized into concrete interaction design strategies for ICH experiences remains underexplored. Existing ICH interactive practices and research broadly exhibit three representative embodied design approaches.

Culturally grounded movement use and semantic feedback. Relevant practices exhibit two noteworthy design features: first, incorporating culturally grounded movement patterns into interaction; and second, establishing semantically coherent feedback between movement and cultural meaning. Guo et al. [12,13], in a Chan Buddhism ICH mixed-reality system, incorporated culturally grounded movements such as “walking with lotus steps” and “pressing palms in meditation” into the interaction, conveying corresponding cultural meanings through feedback such as lotus generation and prayer-seal light effects.

Repetition and continuity of bodily action. Ji et al. [18] shifted ICH craft learning from single-viewing demonstrations into repeated embodied practice through immersive virtual reality, suggesting that bodily action can support learners' understanding and practice of ICH content. This approach is consistent with embodied cognition perspectives suggesting that sustained sensorimotor engagement can contribute to experiential understanding [3,4]. Through structured and continuous bodily action, embodied systems can therefore support users' experience and understanding of cultural content.

The body as a medium for cultural meaning construction. The body serves not only as an input for triggering or displaying content, but can also participate in the construction of cultural meaning. In Nikolakopoulou et al.'s [30] architectural projection-mapping project, participants' walking paths supported spatial navigation through cultural

narratives. Similarly, Hou et al. [16] integrated embodied cues into retrieval mechanisms to facilitate access to tacit cultural knowledge. While these approaches demonstrate how bodily actions can enrich cultural interaction beyond simple trigger mechanisms, their operational focus remains predominantly utilitarian—treating physical movement primarily as a functional interaction mechanism. How the body can actively decode material transformations and abstract aesthetic meanings in traditional craftsmanship remains underexplored.

These existing strategies have primarily been explored in ICH contexts involving established bodily practices—such as Chan meditation movement mapping [12,13], craft-making simulation [18], and martial arts archive retrieval [16]—where embodied forms are already available within the cultural practices. However, when design focus shifts toward abstract aesthetics embedded in material transformation, craft processes, and cultural contexts, existing approaches provide limited guidance on how such aesthetic qualities can be systematically translated into embodied interaction [17] through coordinating bodily action, interactive feedback, and cultural context. Therefore, this study focuses not simply on increasing bodily involvement, but on how designers can construct embodied interactions that connect bodily action with cultural aesthetic meaning. ECMF is proposed in response to this design problem, delineating design translation pathways across three complementary dimensions: movement form, meaning mapping, and tool–material relationships.

2.3 Measurement of Aesthetic Perception in Embodied HCI

Research on aesthetic perception in HCI has gradually expanded from visual aesthetics and its relationship with usability to multidimensional user experience. Tractinsky et al. [35] identified associations between perceived aesthetics and perceived usability. Hassenzahl [14] further distinguished between pragmatic and hedonic attributes, broadening the evaluation of user experience beyond pragmatic qualities alone. However, these classical studies of HCI aesthetics and user experience were not developed specifically for embodied cultural experiences and provide limited coverage of abstract aesthetics and cultural meaning construction. Leder et al.'s [25] cognitive model of aesthetic appreciation further identifies stages, including perceptual analysis, implicit memory integration, explicit classification, cognitive mastery, and evaluation, and highlights the role of associations between prior experience and aesthetic objects in aesthetic understanding. This perspective suggests that aesthetic appreciation involves not only perceptual processing but also the integration of experience and interpretation.

For ICH aesthetics, the perception of abstract aesthetic qualities such as “restraint” and “rusticity” may involve cultural meaning construction, emotional resonance, and prior experience [4], and may therefore be difficult to characterize solely through visual preference or general hedonic-quality measures. Wanzer et al. [36] developed and validated a 22-item Aesthetic Experience Questionnaire (AEQ) using confirmatory factor analysis (CFA), covering dimensions such as emotional engagement, cultural context, sensory perception, cognitive understanding, and flow experience. However, the AEQ was developed primarily for art-appreciation contexts, with items focusing mainly on viewing experiences rather than bodily action and its relation to cultural meaning in embodied interaction. Therefore, assessing aesthetic experience in the specific context of embodied interaction with abstract ICH aesthetics requires attention to experiential dimensions not directly addressed by these measures. Specifically, embodied ICH aesthetic experiences involve not only affective responses and cognitive understanding, but also the interpretation of bodily actions and their relationships with cultural meanings.

Beyond aesthetic perception, presence is another important dimension for evaluating immersive interactive experiences. Witmer et al. [38] examined the factor structure of the Presence Questionnaire (PQ) for assessing presence in immersive environments. In this study, presence is used as a complementary experiential measure to examine differences between interaction conditions alongside ICH aesthetic perception.

Based on these complementary perspectives, the present study develops the ICH Aesthetic Perception Questionnaire (Q2) as a contextualized, exploratory assessment tool. Specifically, Q2 integrates insights from Leder's cognitive model of aesthetic appreciation [25] and Wanzer's Aesthetic Experience Questionnaire (AEQ) [36] to inform the assessment of aesthetic appreciation and experiential responses, while incorporating embodied cognition theory [2,3] and prior research on ICH experience and related cultural meaning measurement [27,40] to address bodily engagement and culturally situated meaning construction. Q2 aims to capture both conventional aesthetic responses and aspects of embodied cultural meaning-making processes and is organized into four dimensions: Emotional Resonance, Cognitive Meaning Construction, Action-based Meaning Perception, and Cultural Meaning Transmission.

3 EMBODIED CULTURAL MOVEMENT FRAMEWORK

3.1 Conceptualization and Design Pathways of Embodied Cultural Movement (ECM)

From Laban's [23] systematic analysis of movement qualities to Klemmer et al.'s [22] synthesis of embodied interaction themes in HCI, prior research has examined the structure and meaning of bodily movement and its role in cognition and interaction from different perspectives. However, how the relationships between bodily movement and abstract cultural aesthetic can be translated into actionable interaction design pathways in ICH contexts remains underexplored. Drawing on the broader principle of Abrahamson et al.'s [1] Embodied Design Framework that bodily movement can support meaning construction, this study proposes the concept of Embodied Cultural Movement (ECM), defined as a design approach that translates abstract aesthetics into embodied interaction by structuring culturally meaningful relationships between bodily action, cultural meaning, and aesthetic experience.

To translate Embodied Cultural Movement into actionable design strategies, this study organizes ECM around three complementary dimensions: how movement unfolds, how bodily action is associated with cultural meaning, and how the body acts through culturally situated tools and materials. Correspondingly, three ECM design pathways are proposed: the Movement-Form Pathway (MFP), the Meaning-Mapping Pathway (MMP), and the Tool-Material Pathway (TMP).

- **Movement-Form Pathway (MFP).** MFP focuses on the formal characteristics of movement, translating dynamic and rhythmic qualities of cultural aesthetics through movement rhythm, spatial extent, and continuity.
- **Meaning-Mapping Pathway (MMP).** MMP focuses on the cultural meanings conveyed through movement, establishing associations by mapping bodily action onto cultural visual features or aesthetic meanings.
- **Tool-Material Pathway (TMP).** TMP focuses on the relationships among body, tools, and materials, connecting embodied interaction with traditional craft practices through tool-material relationships and modes of bodily action.

The three pathways are complementary rather than mutually exclusive and are intended as design-oriented pathways rather than an exhaustive taxonomy of Embodied Cultural Movement. A single embodied interaction may involve multiple pathways while emphasizing one according to its primary design rationale. Together, the three pathways provide a design-oriented framework for translating abstract ICH aesthetics into embodied interaction.

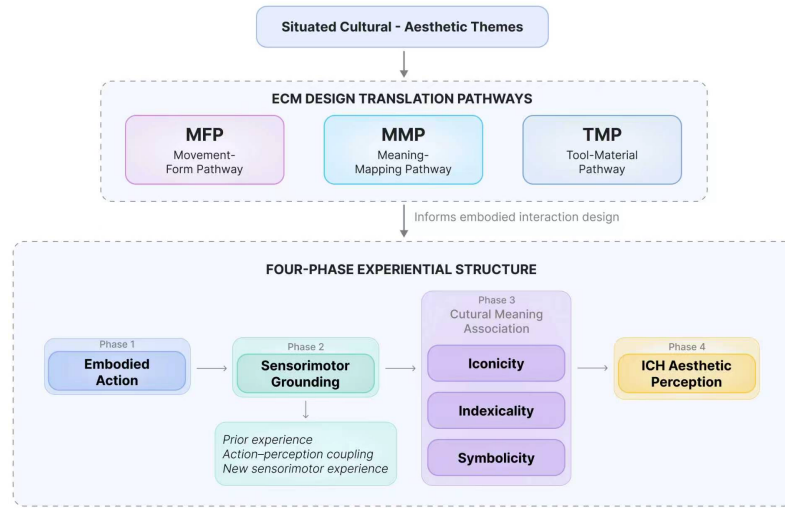


Fig. 2. The Structure of the Embodied Cultural Movement Framework (ECMF)

3.2 Framework Structure

Drawing on embodied cognition theory and Peircean semiotics, this study develops the Embodied Cultural Movement Framework (ECMF) to organize the design translation of abstract ICH aesthetics into embodied interaction. The framework treats bodily action not merely as interactive input, but as a potential experiential medium for connecting sensorimotor experience, cultural meaning, and aesthetic perception.

As shown in Figure 2, ECMF comprises two interrelated components: first, the three ECM design translation pathways consisting of MFP, MMP, and TMP, which serve as complementary design lenses for organizing the translation of abstract aesthetics into embodied interaction; second, a four-phase experiential structure consisting of embodied action, sensorimotor grounding, cultural meaning association, and ICH aesthetic perception, which organizes the experiential relationships among bodily action, sensorimotor experience, cultural meaning, and aesthetic experience. The three design pathways and the four experiential phases are not intended as one-to-one correspondences. The former addresses how abstract aesthetics can be translated into interaction design, whereas the latter conceptualizes the experiential relationships through which bodily engagement may become associated with cultural meaning and aesthetic perception. The four phases are described below.

Phase 1: Embodied Action. Participants perform structured bodily actions designed in relation to the target cultural practice, aesthetic theme, or cultural content, rather than arbitrary physical activity. These actions may draw on movement structures found in traditional craft practice, or establish connections with the target cultural content through movement form, meaning mapping, or tool-material relationships.

Phase 2: Sensorimotor Grounding. Bodily actions associated with cultural practices may engage participants' pre-existing sensorimotor experiences and may also contribute to the formation of sensorimotor associations through action-perception cycles [2,3,11]. Such sensorimotor grounding may provide an experiential basis for further associations between bodily action and cultural content [9,45].

Phase 3: Cultural Meaning Association. Building on sensorimotor experience, bodily action and interactive feedback may establish associations with cultural meaning. Drawing on Peircean semiotics as an analytical lens, ECMF considers

three types of semiotic relations through which such associations may be formed among bodily action, interactive feedback, and cultural meanings or practices: iconicity, indexicality, and symbolicity. These semiotic relations provide an analytical perspective for examining how bodily action and interactive feedback become associated with cultural meaning.

- **Iconicity.** When the form, trajectory, or movement structure of bodily action resembles that of a cultural practice, an iconic relation may be established through formal similarity. For example, simulating movements from a traditional craft practice may create an iconic relation between bodily experience and that cultural practice.
- **Indexicality.** When bodily action directly causes environmental changes or leaves perceptible feedback traces, an indexical relation may be established through causal or contiguous connections between action and outcome. For example, changes in action or applied force may produce corresponding changes in visual feedback, creating a perceptible connection between bodily action and its outcome.
- **Symbolicity.** When bodily action, tool use, or modes of operation are associated with existing cultural conventions or participants' prior cultural experience, a symbolic relation may be established between the action and a specific cultural practice or aesthetic context. For example, the use of a culturally familiar tool in a conventional manner may evoke participants' prior cultural knowledge and experiences, allowing the current action to be interpreted in relation to a broader cultural context.

It should be noted that iconicity, indexicality, and symbolicity represent three semiotic relations rather than sequential steps. These relations may coexist within the same interaction, which may rely primarily on one relation or involve multiple relations depending on movement form, feedback design, and participants' prior cultural experience.

Phase 4: ICH Aesthetic Perception. Through the associations among bodily action, sensorimotor experience, and cultural meaning described in the preceding phases, bodily action and system feedback may serve as experiential cues for perceiving abstract ICH aesthetics. In this way, specific interactive experiences may become associated with culturally situated aesthetic qualities. This phase concerns participants' perception and understanding of abstract ICH aesthetics and their associated cultural meanings. Within ECMF, this experiential dimension is referred to as ICH aesthetic perception.

4 BLUE CALICO INSTALLATION DESIGN

4.1 Why Blue Calico

Blue Calico is a traditional resist-dyeing craft dating back to the Song Dynasty (960–1279) in China. It uses soybean paste as a resist agent and indigo dyeing to produce distinctive white patterns, and has been recognized by the State Council of China as a national-level intangible cultural heritage practice [42]. Blue Calico combines distinctive visual symbols with aesthetic meanings embedded in its craft process [42]. While some cultural meanings can be recognized through patterns and colors, aesthetic dimensions associated with material transformation, craft action, and temporal processes may be difficult to fully convey through visual representation alone. In related work, Yu et al. [43] developed a culture-based knowledge graph for Nantong Blue Calico, providing structured support for interpreting the cultural meanings underlying its craft practices. However, how the abstract aesthetic qualities can be translated into embodied interaction remains underexplored.

4.2 Design Process

4.2.1 Aesthetic Theme Extraction and Design Translation

To ensure that the identification of contextualized aesthetic themes was grounded in relevant cultural materials, this study compiled materials related to Nantong Blue Calico prior to the design process, including academic literature [41–43],

publicly available documentaries, interviews with ICH inheritors, and cultural thematic materials. These materials covered the blue-and-white visual characteristics of Nantong Blue Calico, its hand-dyeing process, traditional pattern symbolism, and material and craft characteristics. The study first conducted an inductive analysis of recurring visual features, craft processes, material transformations, and cultural meanings identified across these materials, and then grouped them according to semantic associations and their potential for embodied interaction design. Following refinement based on expert feedback from an HCI expert and an ICH expert, four contextualized aesthetic themes were selected for subsequent embodied interaction design: Restraint, Rusticity, Vitality, and Temporality. These themes are described below.

- **Restraint.** This theme is primarily reflected in the blue-and-white visual relationship and negative-space characteristics of Blue Calico [41]. Indigo patterns and white negative space complement each other, creating a simple, sparse, and restrained visual quality, which this study characterizes as Restraint.
- **Vitality.** This theme is primarily reflected in the imagery of life and auspicious meanings conveyed through traditional motifs of flowers, plants, birds, and animals [43], as well as aspirations for the continuation of life and well-being embedded in these motifs. This study characterizes these qualities as Vitality.
- **Rusticity.** This theme is primarily associated with the material practices and craft operations involved in handmade production [41]. Physical variations or natural imperfections arising from manual craftwork collectively contribute to an unadorned, rustic authenticity, which this study characterizes as Rusticity.
- **Temporality.** This theme is primarily reflected in the cumulative physical traces and visual changes that gradually unfold across the craft process [42], such as the iterative buildup of dyeing cycles, the emergence of ice-crackle patterns, and gradual material transformation over time. These dynamics capture the passage of time, processual duration, and progressive state transitions, which this study characterizes as Temporality.

These four themes do not represent a comprehensive classification of Blue Calico aesthetics or traditional Chinese aesthetics more broadly. Instead, as shown in Figure 3, they represent contextualized aesthetic themes derived from the collected cultural materials and selected for their relevance to embodied interaction design, including their potential for translation through movement qualities, action–meaning relationships, and body–tool–material interactions. These themes further served as design anchors for operationalizing ECMF, guiding the translation from cultural meanings and material characteristics to embodied actions and interactive feedback.

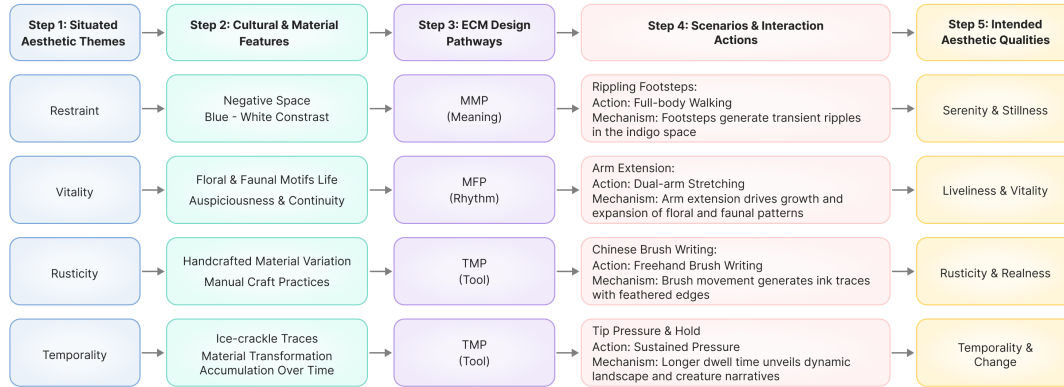


Fig. 3. ECMF-Guided Design Translation from Contextualized Aesthetic Themes to Embodied Interaction

4.2.2 Scene and Interaction Design

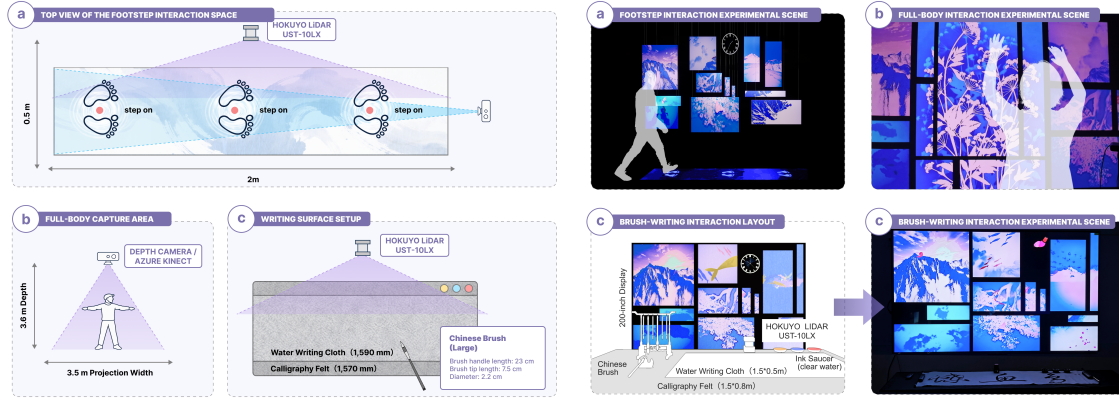


Fig. 4. Interaction Setup and Sensing Layout

In Figure 4, the four contextualized aesthetic themes described above were translated through three interactive scenes: Footstep Ripples, Arm Extension Movement, and Brush Writing. The three scenes differ in the form of bodily involvement, ranging from whole-body locomotion and bilateral arm action to tool-mediated interaction and pressure control. Footstep Ripples uses spatial locomotion as its primary form of interaction; Arm Extension Movement requires participants to actively regulate the amplitude and rhythm of their arm actions; and Brush Writing further introduces culturally familiar tool use and pressure control. Guided by ECMF, the three scenes were designed to primarily emphasize the Meaning-Mapping Pathway (MMP), Movement-Form Pathway (MFP), and Tool-Material Pathway (TMP), respectively, while organizing design correspondences among aesthetic themes, bodily action, and interactive feedback across the dimensions of meaning mapping, movement form, and tool-material relationships.

Scene 1: Footstep Ripples. This scene uses forward locomotion as the primary bodily action, establishing an interaction mapping between physical displacement and visual generation via MMP. As participants step along the floor sensor track, their footsteps trigger localized blue luminous ripples beneath them, concurrently driving visual transitions on the frontal projection from a quiescent, multi-framed layout dominated by negative space into expanding blue-and-white motifs. Rather than replicating literal craft procedures, the system maps participants' bodily intervention onto changes in the blue-and-white visual field, translating the aesthetic theme of Restraint—the dynamic balance between presence and void (liubai)—into an embodied interactive experience.

Scene 2: Arm Extension Movement. The movement form of this scene draws on the large-amplitude arm actions involved in the paste-scraping process of Blue Calico production. Through MFP, changes in arm amplitude and continuity are linked to corresponding changes in visual feedback. As participants extend their arms, flower motifs gradually grow in response to the amplitude of the movement, establishing a design connection between the progressive extension of the arms and the life-related meanings embedded in traditional motifs.

Scene 3: Brush Writing. Drawing on the craft operations and material characteristics of pattern drawing and paste application in Blue Calico production, this scene employs TMP to introduce the brush as a culturally familiar tool associated with traditional Chinese craft practices. The interaction connects bodily action, tool use, and feedback with the aesthetic themes of Rusticity and Temporality. Participants control changes in brush strokes through applied pressure,

directly evoking the manual craft practice and material qualities associated with Rusticity. Meanwhile, sustained pressure and dwell time at a specific position introduce a temporal accumulation process. As time accumulates, segmented visual narratives are progressively revealed and animated within the projected canvas, making the passage of time and material transformation associated with Temporality perceptible through embodied feedback.

4.3 Technical Implementation

The installation used a 3840×2160 , 60 Hz projector as the visual display interface. A custom pressure-sensing floor sensor array was used for footstep detection, an RGB-D sensor tracked upper-limb movement for gesture interaction, and a radar sensor detected tool movement within the tabletop (1920×1080) interaction area. A four-channel surround spatial audio system provided auditory feedback, and a custom pressure-sensitive brush enabled brush pressure control. The overall spatial configuration and hardware setup are shown in Figure 5.

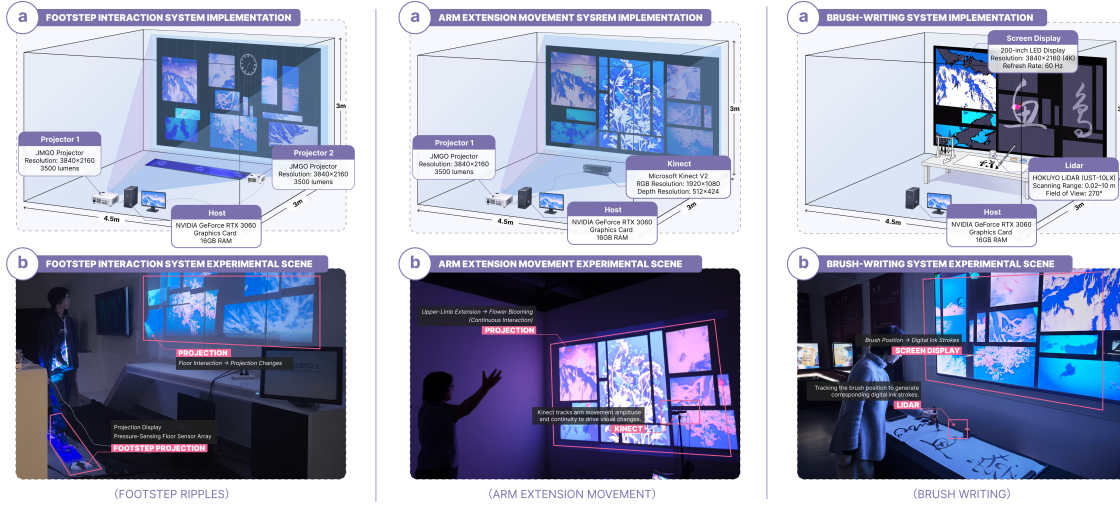


Fig. 5. Spatial Setup and Hardware Setup

5 EVALUATION METHOD

5.1 Measurement Dimensions and Instruments

To examine experiential differences associated with ECMF-guided embodied interaction, this study adopted a convergent mixed-methods approach that integrated quantitative questionnaires with qualitative inquiries. Two primary experiential dimensions were assessed: presence and ICH aesthetic perception. Presence was measured using the Presence Questionnaire (PQ), while ICH aesthetic perception was assessed using the contextualized ICH Aesthetic Perception Questionnaire (Q2) developed in this study.

Presence (PQ). This study used the Presence Questionnaire (PQ) developed by Witmer et al. [38] to assess presence in the immersive environment. The overall presence score served as the primary outcome measure, while three subscales—Involvement/Control (INV/C), Natural (NATRL), and Interface Quality (IFQUAL)—were included in exploratory analyses to examine differences in presence between the EC and LEC conditions.

ICH Aesthetic Perception (Q2). Because existing scales do not directly cover the Blue Calico ICH aesthetic context examined in this study, a contextualized ICH Aesthetic Perception Questionnaire (Q2) was developed as an exploratory measure. The item development was based on the aesthetic themes identified in the design process and informed by existing perspectives on aesthetic experience and embodied cognition. The initial items were reviewed by two experts, one specializing in intangible cultural heritage and the other in HCI, and were revised based on their feedback regarding conceptual relevance, wording clarity, and contextual appropriateness. The revised questionnaire was subsequently administered to an independent pilot sample ($n = 26$) to examine item clarity and internal consistency. The final Q2 consisted of 22 items organized across four dimensions: Emotional Resonance ($\alpha = .789$), Cognitive Meaning Construction ($\alpha = 0.752$), Action-based Meaning Perception ($\alpha = .868$), and Cultural Meaning Transmission ($\alpha = 0.826$). All items were rated on a 7-point scale.

In addition, the Immersive Tendencies Questionnaire (ITQ) was used to characterize participants' baseline immersive tendencies rather than as an outcome measure of presence. Qualitative data were collected through semi-structured interviews and analyzed using thematic analysis [6]. Two coders independently coded the data, and inter-coder agreement was assessed using Cohen's k ($k = .83$).

5.2 Participants

Participants were recruited based on their responses to a pre-study questionnaire (Q1), which assessed their prior knowledge of and familiarity with traditional Chinese culture. This recruitment approach was adopted to provide participants with sufficient contextual background for engaging with the culturally situated aesthetic content examined in this study. A total of 25 participants (aged 21–30 years) was recruited. Three participants were excluded because equipment malfunctions prevented them from completing the experimental procedure, resulting in a final sample of 22 participants ($n = 22$), including 11 females and 11 males. Participants came from a range of disciplinary backgrounds, including arts, social sciences, and science and engineering. The study protocol and recruitment procedure were approved by the institutional ethics review board of the authors' institution prior to data collection. All participants provided written informed consent before the experiment.

5.3 Experimental Design

A within-subjects counterbalanced crossover design [37] was employed to compare two interaction conditions: the Embodied Condition (EC) and the Low-Embodiment Condition (LEC). Both conditions presented the same cultural content, visual elements, and interaction scenes, while differing in how participants physically interacted with the content. In the EC condition, participants interacted through the bodily actions designed according to ECMF, including walking, arm extension, and brush manipulation, receiving corresponding multimodal feedback. In the LEC condition, participants interacted with the same cultural and visual content using a wireless remote controller. Each interaction condition consisted of three scenes, and each scene lasted approximately 15 minutes, including questionnaire completion. The mappings between controller inputs and visual outcomes were designed to preserve the same interaction outcomes as in the EC condition. Specifically, the scroll wheel controlled the growth of visual motifs, a single button press triggered the footstep ripples, and a sustained button press triggered the progressive unveiling and animation of the multi-framed visual narrative. This comparison was designed to examine whether ECMF-guided embodied interaction was associated with differences in presence and ICH aesthetic perception compared with a low-embodiment interaction condition.

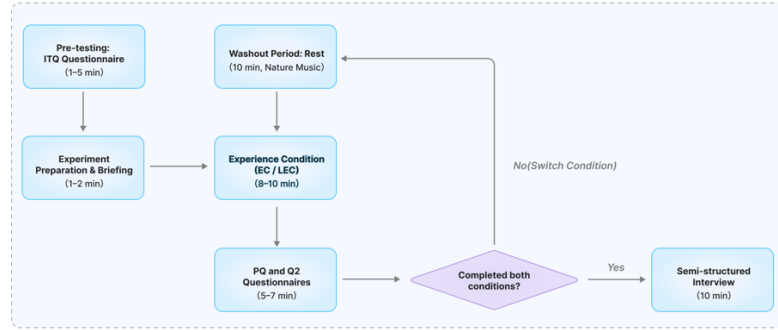


Fig. 6. Within-subjects crossover experimental procedure

6 RESULTS

Following the convergent mixed-methods design, quantitative ratings and qualitative findings are reported below to provide a complementary view of participants' experiential responses.

6.1 Quantitative Analysis

Paired-samples t-tests were used for the primary comparisons between the EC and LEC conditions, with Cohen's d_z reported as the effect size for paired comparisons. The overall PQ score was treated as a primary outcome, whereas analyses of PQ subscales and Q2 dimensions were considered exploratory and corrected for multiple comparisons. For the four Q2 dimensions, Bonferroni correction was applied, resulting in the significance threshold $\alpha = .0125$. For the three PQ subscales (INV/C, NATRL, IFQUAL), Bonferroni correction was similarly applied ($\alpha = .0167$). Potential order effects were further examined by comparing EC–LEC difference scores between the two condition-order groups using independent-samples t-tests.

Presence. Overall presence was significantly higher in the EC condition than in the LEC condition, with a mean difference of 0.45 points (95% CI [0.21, 0.69]), $t(21) = 3.94, p < .001, d_z = 0.84$. Among the PQ subscales, Natural (NATRL) showed a mean difference of 1.09 points (95% CI [0.50, 1.69]), $t(21) = 3.81, p < .001, d_z = 0.81$, while Involvement/Control (INV/C) showed a mean difference of 0.73 points (95% CI [0.41, 1.04]), $t(21) = 4.81, p < .001, d_z = 1.02$. Both differences remained significant at the Bonferroni-adjusted significance level ($\alpha = .0167$). No significant difference was observed for Interface Quality (IFQUAL) ($p > .05$). These results indicate that the difference in presence between the two conditions was primarily reflected in Natural and Involvement/Control.

ICH Aesthetic Perception. As shown in Table 1, scores for all four Q2 dimensions were higher in the EC condition than in the LEC condition. Emotional Resonance showed a mean difference of 1.11 points (95% CI [0.89, 1.32]), $t(21) = 10.84, p < .001, d_z = 2.32$. Cognitive Meaning Construction showed a mean difference of 0.91 points (95% CI [0.60, 1.22]), $t(21) = 6.09, p < .001, d_z = 1.30$. For Action-based Meaning Perception, the mean difference was 0.65 points (95% CI [0.18, 1.12]), $t(21) = 2.85, p = .010, d_z = 0.61$. Cultural Meaning Transmission showed a mean difference of 0.61 points (95% CI [0.22, 1.00]), $t(21) = 3.28, p = .004, d_z = 0.70$. All four differences remained significant at the Bonferroni-adjusted threshold of $\alpha = .0125$.

Order Effects. Potential order effects were examined by comparing EC–LEC difference scores between the two condition-order groups. No differences remained significant after Bonferroni correction. Specifically, Emotional Resonance ($p = .146$), Action-based Meaning Perception ($p = .181$), and Cultural Meaning Transmission ($p = .253$)

showed no significant differences between the two condition-order groups. For Cognitive Meaning Construction, the EC→LEC difference score was 0.51 (SD = 0.55) for the EC→LEC group ($n = 11$) and 1.18 (SD = 0.68) for the LEC→EC group ($n = 11$). The difference was significant before correction, $t(20) = -2.47$, $p = .022$, 95% CI [-1.24, -0.11], Cohen's $d = 1.07$. However, this difference did not remain significant after Bonferroni correction ($\alpha = .0125$).

Table 1. Descriptive Statistics for ICH aesthetic perception (Q2)

Dimension	EC: M (SD)	LEC: M (SD)	ΔM (EC-LEC)	95% CI	$t(21)$	p	d_z
Emotional Resonance (ER)	5.25 (0.58)	4.14 (0.43)	1.11	[0.89, 1.32]	10.84	< .001	2.32
Cognitive Meaning Construction (CM)	4.96 (1.06)	4.05 (0.72)	0.91	[0.60, 1.22]	6.09	< .001	1.30
Action-based Meaning Perception (AM)	5.52 (0.90)	4.88 (1.09)	0.65	[0.18, 1.12]	2.85	.010	0.61
Cultural Meaning Transmission (CMT)	4.90 (1.06)	4.29 (0.99)	0.61	[0.22, 1.00]	3.28	.004	0.70

Note: $N = 22$. Scores are on a 7-point Likert scale. Values are reported as Mean (Standard Deviation). Significance threshold after Bonferroni correction is $\alpha = .0125$ ($0.05 / 4$). d_z denotes Cohen's d_z for paired-samples design.

6.2 Qualitative Findings

As shown in Table 2, thematic analysis identified three core themes, with good inter-coder agreement (Cohen's $\kappa = .82$), covering cultural association and contextual meaning construction, bodily perception and connection with prior experience, and the role of cultural tools on bodily operations.

Table 2. Qualitative Themes and Operational Definitions

Theme	Data Source	Key Codes	Keywords / Operational Definitions
Cultural Associations and Contextual Meaning Construction	Interview	Scene-based associations; Memory-based associations	Connections between the interaction and imagined cultural/natural scenes, prior writing or artistic practices, and personal memories
Bodily Perception and Connections with Prior Experience	Observation; Interview	Pressure sensation; Pausing/resistance sensation; Movement sensation; Action–visual feedback coupling; Friction/discomfort	Connections between bodily sensations during interaction, action–feedback relationships, and participants' prior bodily or practice-based experiences.
Cultural Familiarity in Tool Use	Observation; Interview	Calligraphy-like grip; Attempts to use the brush; Attention to tool manipulation	Ways in which the culturally familiar form and use of a tool guided bodily action, interaction choices, and attention

Theme 1: Cultural Associations and Contextual Meaning Construction. Some participants connected their interactive experiences with specific scenes, prior experiences, or cultural contexts. These connections primarily took two forms: scene-based associations and memory-based associations. Scene-based associations ($n = 11$) often involved imaginings of natural environments evoked by audiovisual and bodily experiences. For example, P12 described the experience as evoking “a lake and reeds” and “sunlight shimmering on the water.” Memory-based associations ($n = 6$)

more often involved participants' existing writing, artistic practice, or life experiences. For example, during brush interaction, P20 associated the experience with previously copying ancient calligraphy models. These two forms of association were not mutually exclusive, and the same participant could establish multiple forms of meaning connection. However, such cultural associations did not appear for all participants. Some focused more on task completion and interaction operation. For example, P22 reported no particular association when asked. These differences suggest that cultural association may emerge as one form of meaning construction during embodied experience, while its formation may depend on participants' prior experience and mode of engagement with the interaction.

Theme 2: Bodily Perception and Connections with Prior Experience. Some participants connected the tactile and kinesthetic sensations during brush manipulation with their prior writing experience or bodily memory, including pressure sensation ($n = 3$), pauses or resistance during brush movement ($n = 2$), and movement sensation ($n = 2$). For example, P13 described the sensation of pausing the brush as familiar and evocative of prior experience. These accounts suggest that subtle bodily feedback during tool use may serve as a cue connecting the current interaction with prior bodily experience. Meanwhile, six participants reported friction or discomfort arising from direct contact with the device, indicating that stronger bodily feedback is not necessarily associated with a better experience, and its compatibility with the cultural practice and mode of operation may be more important. Participants also noticed the coupling between bodily action and visual feedback. For example, P15 described how, when extending the arms, "the image seemed to open up with the movement." This experience corresponded to the design of the Arm Extension Movement scene, in which movement amplitude and continuity drove changes in visual feedback. In addition, some participants continued writing or drawing after the interview, indicating sustained engagement with the tool-based activity. Taken together, these observations reveal connections among bodily perception, tool use, and prior experience.

Theme 3: Cultural Familiarity in Tool Use. Interview data and behavioral observation indicated that the form of the brush and participants' familiarity with its use influenced how they interacted with the system. Twenty of 22 participants spontaneously adopted a grip similar to conventional brush-writing practices, including five participants who had no prior experience using a brush. Notably, in the LEC condition, five participants attempted to pick up the visible but inactive brush even though they had been instructed to use the remote controller. These behaviors suggest that the culturally familiar form and conventional use of the brush provided cues for how participants interpreted the available interaction possibilities and selected forms of action. A related pattern was observed in participants' allocation of attention. During brush interaction, 13 participants reported focusing primarily on the act of drawing rather than waiting for visual feedback on the screen, and three reported paying relatively little attention to changes in the visual display. For some participants, therefore, the brush interaction shifted attention from the visual outcome toward the bodily process of tool use, making the act of manipulation itself an important part of the experience. Together, these observations reveal how tool form, culturally familiar modes of use, and bodily action became intertwined during interaction.

7 DISCUSSION

Using Nantong Blue Calico as a case, this study examined how contextualized aesthetic themes such as 'Restraint, Rusticity, Vitality, and Temporality' can be translated into embodied interactive experiences through bodily action, cultural meaning association, and tool-material relationships. Compared with the LEC condition, the EC condition resulted in higher overall presence and higher scores across the four dimensions of ICH aesthetic perception. These findings suggest that ECMF-guided embodied interaction was associated with a distinct mode of engaging with abstract ICH aesthetics, in which bodily experience became connected with cultural meaning construction, compared with low-embodiment interaction.

7.1 Interpreting ECMF-guided Embodied Interaction

The quantitative ratings and qualitative themes provide complementary insights into how ECMF-guided interaction may support engagement with abstract ICH aesthetics. The qualitative findings revealed cultural associations, connections between bodily perception and prior experience, and culturally familiar patterns of tool use. These findings suggest that ECMF-guided interaction may organize embodied experiences through which participants can engage with cultural meanings and aesthetic qualities.

These observations are broadly consistent with the experiential relationships described in ECMF among bodily action, sensorimotor experience, cultural meaning, and ICH aesthetic perception. However, this study evaluated experiential outcomes rather than independently measuring or manipulating intermediate processes. Therefore, these findings provide interpretive evidence for participants' experiences rather than the specific mechanisms or sequential relationships proposed among ECMF phases.

Similarly, the Movement-Form Pathway (MFP), Meaning-Mapping Pathway (MMP), and Tool-Material Pathway (TMP) jointly informed the prototype design but were not tested as independent experimental variables. Therefore, the current findings reflect differences associated with the integrated ECMF-guided design rather than the independent contribution of individual pathways. In this sense, the study should be understood as a framework-informed design evaluation rather than an examination of the independent effects of individual ECMF pathways.

7.2 Differences across Dimensions of ICH Aesthetic Perception

All four dimensions of ICH aesthetic perception showed higher scores in the EC condition than in the LEC condition, although the magnitude of the differences varied across dimensions. The largest differences were observed for Emotional Resonance ($d_z = 2.32$) and Cognitive Meaning Construction ($d_z = 1.30$), whereas smaller differences were observed for Action-based Meaning Perception ($d_z = 0.61$) and Cultural Meaning Transmission ($d_z = 0.70$). These differences in effect size suggest that the four dimensions may capture different aspects of participants' experiences that respond differently to the contrast between EC and LEC conditions.

Qualitative reflections from Theme 1 (e.g., spontaneous associations with natural scenes and calligraphy practice) help explain this pronounced contrast, indicating that bodily engagement may have supported participants' connections with pre-existing affective and cultural associations. Conversely, the more moderate effect size observed for Action-based Meaning Perception can be understood through Theme 2 and Theme 3: as participants focused on physical brush manipulation, their attention frequently shifted toward the immediate act of drawing rather than simultaneously monitoring on-screen feedback, which may have limited their explicit awareness of this action-feedback correspondence.

The result for Cognitive Meaning Construction should nevertheless be interpreted with additional caution. The comparison of EC-LEC difference scores between the two condition-order groups yielded $p = .022$ and Cohen's $d = 1.07$. Although this difference did not reach the Bonferroni-adjusted significance threshold of $\alpha = .0125$, its magnitude suggests a potential order-related pattern that warrants further examination. Participants who experienced one condition first may have developed different interpretive perspectives on the cultural content that influenced their subsequent experience of the other condition. Given the current sample size and the lack of statistical significance after correction, however, this possibility should be treated as a tentative interpretation rather than evidence of a reliable carryover effect.

More broadly, differences in effect size indicate the relative strength of the observed differences between the two experimental conditions; they do not establish a temporal ordering among different aspects of aesthetic experience. Because the present study measured ICH aesthetic perception retrospectively after each condition, it did not further examine how these experiential dimensions emerged, changed, or interacted during the course of interaction. Future studies could

incorporate continuous experience sampling, process-oriented interviews, or other dynamic measures to examine how these dimensions change and relate to one another during embodied interaction.

7.3 From Cultural Representation to Bodily Participation

The findings of this study suggest a broader design perspective for experiencing abstract ICH aesthetics in digital experiences: cultural content need not be conveyed only through visual representation, but can also be incorporated into bodily participation. In ECMF, abstract aesthetic qualities are translated into embodied interaction through movement form, meaning mapping, and tool–material relationships. From this perspective, the body functions not merely as a means of triggering digital content, but as part of the process through which cultural meanings and aesthetic qualities may be encountered.

This perspective is particularly relevant to abstract ICH aesthetics that cannot be readily translated through the direct reproduction of existing bodily actions. In such cases, the design goal need not be to reproduce traditional actions as faithfully as possible, but rather to construct a coherent relationship among bodily action, interactive feedback, and the target aesthetic quality. Footstep Ripples, for example, does not reproduce a bodily action from the traditional Blue Calico production process. Instead, it establishes a correspondence between bodily intervention and changes in the blue-and-white visual field, allowing the spatial relationship associated with Restraint to be explored through bodily participation. Brush Writing takes a different approach by using a culturally familiar tool and mode of operation to connect bodily action with the craft context through tool–material relationships.

ECMF therefore does not prescribe a fixed set of bodily actions for representing abstract ICH aesthetics. Rather, it provides a design-oriented way of organizing relationships among bodily action, interactive feedback, cultural associations, and target aesthetic qualities.

7.4 Design Implications

Based on the quantitative and qualitative findings, this study identifies four design implications for embodied interaction with abstract ICH aesthetics.

DI#1: Embed Cultural Meaning in the Interactive Process, Not Only in the Final Representation. Cultural meaning may be embedded in the ongoing relationship between bodily action and system feedback rather than conveyed only through the final visual representation. The cultural associations, action–feedback relationships, and tool-use behaviors observed in this study illustrate how bodily action can support users’ engagement with cultural meaning during interaction. Designers may therefore organize cultural visual features, material transformations, or aesthetic meanings through continuous action–feedback relationships, allowing cultural content to be encountered through bodily action rather than read only from a final visual outcome. From this perspective, feedback design can shift from an “action–result” structure toward a continuous relationship among bodily action, feedback, and cultural meaning.

DI#2: Align Embodied Interaction with Material and Aesthetic Characteristics of the Craft. Building on the Movement-Form and Tool–Material Pathways of ECMF, bodily feedback should be designed in relation to the aesthetic qualities of the target craft rather than simply intensified. Participants reported subtle bodily sensations associated with pressure, pauses or resistance, and movement, while some also reported friction or discomfort. These findings suggest that the design of bodily feedback should consider its compatibility with the material qualities, operational rhythms, and aesthetic context of the target craft. For traditional crafts characterized by restraint, subtlety, or slow processes, moderate and process-oriented feedback may be more appropriate than intense sensory stimulation.

DI#3: Consider Cultural Familiarity in Interactive Tool Selection and Use. The selection of interactive tools should consider not only general usability but also their cultural familiarity and relationship to the target cultural practice. Participants' spontaneous ways of holding and using the brush suggest that culturally familiar tools can provide cues for how users understand and perform bodily actions. Accordingly, the form and conventional use of an interactive tool may serve as design resources for connecting bodily action with prior cultural experience and the broader cultural context.

DI#4: Account for Differences in Prior Cultural Experience. Participants differed in the extent to which they formed cultural associations or connected the interaction with prior experiences. Embodied translation of ICH aesthetics should therefore not assume that all users share the same cultural background. For users with relevant prior experience, familiar actions, tools, or cultural imagery may provide resources for establishing experiential connections. For users with less relevant experience, appropriate visual, auditory, or contextual cues may provide cultural scaffolding that helps connect bodily action with cultural meaning.

7.5 Limitations and Future Research

This study has several limitations that also suggest directions for future research. First, the final sample size was relatively small ($n = 22$), and the stability and generalizability of the findings should be examined with larger and more diverse samples. In particular, the relatively large effect observed for Emotional Resonance ($d_z = 2.32$) should be interpreted cautiously, as effect estimates from small samples may be sensitive to sample characteristics. Future studies with larger and more diverse samples are needed to further examine the robustness and generalizability of these effects.

Second, the comparison between the EC and LEC conditions did not isolate a single component of embodied interaction. The EC condition combined whole-body action, culturally situated tool use, and richer sensorimotor involvement, whereas the LEC condition used a wireless handheld controller to interact with the same cultural content. Therefore, the observed differences should be interpreted as differences between two integrated interaction conditions rather than as the independent effect of bodily action, tool use, or any single ECMF pathway. Future studies could use ablation studies or factorial designs to manipulate these components separately and examine the individual and combined contributions of MFP, MMP, and TMP.

Third, the within-subjects crossover design may have introduced learning or carryover effects between conditions. For Cognitive Meaning Construction, the comparison of EC–LEC difference scores between the two condition-order groups yielded $p = .022$ and Cohen's $d = 1.07$. Although this difference did not reach the Bonferroni-adjusted significance threshold of $\alpha = .0125$, its magnitude suggests that a potential order-related pattern cannot be ruled out. Experience with the first condition may have influenced how some participants understood or evaluated the cultural content in the subsequent condition. Future studies could use between-subjects designs, longer intervals or washout tasks between conditions, or statistical models that explicitly account for order and carryover effects.

Fourth, the cultural specificity of the present case limits the generalizability of the design implications. The four design implications identified in this study were derived from a single Blue Calico case and should therefore be further examined across other ICH practices and cultural contexts. In addition, participants were recruited based on a pre-study questionnaire assessing prior knowledge of and familiarity with traditional Chinese culture, and the sample therefore does not fully represent users with diverse levels of cultural familiarity. The roles of culturally familiar tools, bodily actions, and cultural associations may differ across users with different cultural backgrounds or prior experience, which warrants further investigation through cross-cultural studies.

Finally, Q2 was developed as a contextualized exploratory measure for embodied interaction with abstract ICH aesthetics. Although its four dimensions demonstrated acceptable to good internal consistency in the independent pilot

sample ($n = 26$), the current sample size was insufficient for establishing its factor structure through exploratory factor analysis (EFA) or confirmatory factor analysis (CFA). Q2 should therefore be interpreted as a contextualized assessment tool developed for this study rather than as a validated general-purpose scale. Future research should evaluate its factor structure, construct validity, and applicability across different ICH contexts using larger, independent samples.

8 CONCLUSION

This study addressed the design challenge of translating abstract ICH aesthetics into embodied interaction by proposing the Embodied Cultural Movement Framework (ECMF). ECMF provides three design pathways—the Movement-Form Pathway (MFP), Meaning-Mapping Pathway (MMP), and Tool-Material Pathway (TMP)—and a four-phase experiential structure comprising Embodied Action, Sensorimotor Grounding, Cultural Meaning Association, and ICH Aesthetic Perception. Through a case study of Nantong Blue Calico, the comparison between the EC and LEC conditions showed higher overall presence and higher scores across the four dimensions of ICH aesthetic perception in the EC condition, while the qualitative findings further revealed cultural associations, connections between bodily perception and prior experience, and culturally familiar patterns of tool use. These findings provide preliminary design and empirical support for translating abstract ICH aesthetics that cannot be readily represented through existing bodily actions into embodied interaction, and provide an initial basis for future research on embodied interaction with abstract ICH aesthetics.

References

- [1] Abrahamson, D. (2014). Building educational activities for understanding: An elaboration on the embodied-design framework and its epistemic grounds. *International Journal of Child-Computer Interaction*, 2(1), 1–16. <https://doi.org/10.1016/j.ijcci.2014.07.002>
- [2] Barsalou, L. W. (1999). Perceptual symbol systems. *Behavioral and Brain Sciences*, 22(4), 577–660. <https://doi.org/10.1017/S0140525X99002149>
- [3] Barsalou, L. W. (2003). Situated simulation in the human conceptual system. *Language and Cognitive Processes*, 18(5-6), 513–562. <https://doi.org/10.1080/01690960344000026>
- [4] Barsalou, L. W. (2009). Simulation, situated conceptualization, and prediction. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1521), 1281–1289. <https://doi.org/10.1098/rstb.2008.0319>
- [5] Bertrand, S., Vassiliadi, M., Zikas, P., Geronikolakis, E., & Papagiannakis, G. (2021). From readership to usership: Communicating heritage digitally through presence, embodiment and aesthetic experience. *Frontiers in Communication*, 6, 676446. <https://doi.org/10.3389/fcomm.2021.676446>
- [6] Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [7] Clark, A. (1997). *Being There: Putting Brain, Body, and World Together Again*. MIT Press.
- [8] Dhanda, A., Henderson, J., Hermon, S., & Bevan, A. (2019). Recreating cultural heritage environments for VR. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLII-2/W9, 305–310. <https://doi.org/10.5194/isprs-archives-XLII-2-W9-305-2019>
- [9] Engelkamp, J., & Zimmer, H. D. (1994). *Human Memory: A Multi-Modal Approach*. Hogrefe & Huber Publishers.
- [10] Foglia, L., & Wilson, R. A. (2013). Embodied cognition. *WIREs Cognitive Science*, 4(3), 319–325. <https://doi.org/10.1002/wcs.1226>
- [11] Glenberg, A. M., & Gallese, V. (2012). Action-based language: A theory of language acquisition, comprehension, and production. *Cortex*, 48(7), 905–922. <https://doi.org/10.1016/j.cortex.2011.04.010>
- [12] W. Guo, Z. Chen, G. Sun and H. Wang, "'Heart Flows with Zen': Exploring Multi-Modal Mixed Reality to Promote the Inheritance and Experience of Cultural Heritage," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 11, pp. 9548-9558, Nov. 2025, doi: 10.1109/TVCG.2025.3616750.
- [13] Guo, W. C., Zhao, W. B., Sun, G. Y., et al. (2024). Enhancing the digital inheritance and embodied experience of Zen based on multimodal mixed reality system. In *ACM SIGGRAPH 2024 Posters* (pp. 1–2). ACM.
- [14] Hassenzahl, M. (2004). The interplay of beauty, goodness, and usability in interactive products. *Human-Computer Interaction*, 19(4), 319–349. https://doi.org/10.1207/s15327051hci1904_2
- [15] Yumeng Hou, Sarah Kenderdine, Davide Picca, Mattia Egloff, and Alessandro Adamou. 2022. Digitizing Intangible Cultural Heritage Embodied: State of the Art. *J. Comput. Cult. Herit.* 15, 3, Article 55 (September 2022), 20 pages. <https://doi.org/10.1145/3494837>
- [16] Hou, Y., Seydou, F., & Kenderdine, S. (2023). Unlocking a multimodal archive of Southern Chinese martial arts through embodied cues. *Journal of Documentation*, 79(3), 684–706. <https://doi.org/10.1108/JD-01-2022-0027>
- [17] Hu, H., Li, Q., & Cao, X. (2022). METH research on design paradigm of cultural heritage based on embodied cognition. *International Journal of Human-Computer Interaction*, 41(3), 1860–1871. <https://doi.org/10.1080/10447318.2022.2087268>

- [18] Ji, Y., Zhong, S., Zhang, B., & Sean, C. (2021). Constructing embodied interaction of intangible cultural heritage course through immersive virtual reality. In Proceedings of the Ninth International Symposium of Chinese CHI (pp. 1–11). ACM. <https://doi.org/10.1145/3490355.3490373>
- [19] Kenderdine, S. (2013). "Pure Land": Inhabiting the Mogao Caves at Dunhuang. Curator: The Museum Journal, 56(2), 199–218. <https://doi.org/10.1111/cura.12020>
- [20] Kenderdine, S. (2016). Embodiment, entanglement, and immersion in digital cultural heritage. In S. Schreibman et al. (Eds.), A New Companion to Digital Humanities (pp. 22–41). John Wiley & Sons.
- [21] Kirsh, D. (2013). Embodied cognition and the magical future of interaction design. ACM Transactions on Computer-Human Interaction, 20(1), 1–30. <https://doi.org/10.1145/2442106.2442109>
- [22] Klemmer, S. R., Hartmann, B., & Takayama, L. (2006). How bodies matter: Five themes for interaction design. In Proceedings of the 6th Conference on Designing Interactive Systems (DIS 2006) (pp. 140–149). ACM. <https://doi.org/10.1145/1142405.1142429>
- [23] Laban, R. (1950). The Mastery of Movement. Plays Inc.
- [24] Lakoff, G., & Johnson, M. (1999). Philosophy in the Flesh: The Embodied Mind and Its Challenge to Western Thought. Basic Books.
- [25] Leder, H., Belke, B., Oeberst, A., & Augustin, D. (2004). A model of aesthetic appreciation and aesthetic judgments. British Journal of Psychology, 95(4), 489–508. <https://doi.org/10.1348/0007126042369811>
- [26] Liu, L., Zhang, Y., Meng, J., Lian, X., Wei, J., & Wang, J. (2023). An analysis of the "blank space" in freehand flower and bird painting from an aesthetic perspective. Journal of Arts and Cultural Studies, 2(2), 1–11. <https://doi.org/10.23112/acs23080701>
- [27] Liu, Y., Li, Y., Tao, W., & Wang, Q. (2024). Measuring intangible cultural heritage image: A scale development. PLOS ONE, 19(6), e0299088. <https://doi.org/10.1371/journal.pone.0299088>
- [28] Lu, Y., Liu, Z., Yan, X., & Yan, S. (2021). Virtual narrative and creative methods of oral performance in China intangible cultural heritage. In SIGGRAPH Asia 2021 XR (pp. 1–2). ACM. <https://doi.org/10.1145/3478514.3487623>
- [29] Musicco, A., Rossi, N., & Verdoscia, C. (2023). Accuracy evaluation of smartphone-based videogrammetry for cultural heritage documentation. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLVIII-M-2-2023, 1119–1126. <https://doi.org/10.5194/isprs-archives-XLVIII-M-2-2023-1119-2023>
- [30] Nikolakopoulou, V., et al. (2022). Conveying intangible cultural heritage in museums with interactive storytelling and projection mapping. Heritage, 5(2), 1024–1049. <https://doi.org/10.3390/heritage5020056>
- [31] Patrucco, G., et al. (2023). Digital replicas of British Museum artefacts. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLVIII-M-2-2023, 1173–1180. <https://doi.org/10.5194/isprs-archives-XLVIII-M-2-2023-1173-2023>
- [32] Peirce, C. S. (1931–1958). Collected Papers of Charles Sanders Peirce (Vols. 1–8). Harvard University Press.
- [33] Phan, N., Le, P. H., & Tran, H. V. V. (2024). Digitizing heritage: Evaluating the effectiveness of the PhotoScan method in artifact preservation at Lam Dong Museum, Vietnam. International Journal of Scientific Research in Civil Engineering, 8(4), 102–111. <https://doi.org/10.32628/ijsrcce248411>
- [34] Polanyi, M. (1966). The Tacit Dimension. Doubleday.
- [35] Tractinsky, N., Katz, A. S., & Ikar, D. (2000). What is beautiful is usable. Interacting with Computers, 13(2), 127–145. [https://doi.org/10.1016/S0953-5438\(00\)00031-X](https://doi.org/10.1016/S0953-5438(00)00031-X)
- [36] Wanzer, D. L., et al. (2020). Aesthetic experience and its measurement. Psychology of Aesthetics, Creativity, and the Arts, 14(2), 236–249. <https://doi.org/10.1037/aca0000203>
- [37] Lazar, J., Feng, J. H., & Hochheiser, H. (2017). Research Methods in Human-Computer Interaction (2nd ed.). Morgan Kaufmann.
- [38] Witmer, B. G., Jerome, C. J., & Singer, M. J. (2005). The factor structure of the presence questionnaire. Presence: Teleoperators and Virtual Environments, 14(3), 298–312. <https://doi.org/10.1162/105474605323384654>
- [39] Wu, S., et al. (2021). On the aesthetic influence of the Song Dynasty literati painting on porcelain aesthetics. International Journal of Frontiers in Sociology, 3(9). <https://doi.org/10.25236/IJFS.2021.030904>
- [40] Yan, X., et al. (2024). Ritualized ICH experience scale for tourism contexts. Journal of Destination Marketing & Management, 31, 100843. <https://doi.org/10.1016/j.jdmm.2023.100843>
- [41] Yan, Y. (2024). Aesthetic flowing between islands: Migrating loom, traded cotton, mountain of dye. Journal of Material and Intangible Culture, 13(2). <https://doi.org/10.21463/jmic.2024.13.2.02>
- [42] Yin, C. (2024). *An examination of the symbolic significance and cultural inheritance of ICH in China: A case study of Nantong blue calico* (Doctoral thesis, Nanyang Technological University). <https://doi.org/10.32657/10356/174503>
- [43] Yu, X., Shen, M., & Zhang, L. (2026). Knowledge system based on cultural repository of Nantong blue calico patterns. *Digital Applications in Archaeology and Cultural Heritage*, 40, e00500. <https://doi.org/10.1016/j.daach.2026.e00500>
- [44] Zhang, L., Wang, Y., Tang, Z., Liu, X., & Zhang, M. (2023). A virtual experience system of bamboo weaving for sustainable research on intangible cultural heritage based on VR technology. Sustainability, 15(4), 3134. <https://doi.org/10.3390/su15043134>
- [45] Zimmer, H. D. (2001). Why do actions speak louder than words? In H. D. Zimmer et al. (Eds.), Memory for Action. Oxford University Press.

APPENDIX A. ITEMS OF THE ICH AESTHETIC PERCEPTION QUESTIONNAIRE (Q2)

The ICH Aesthetic Perception Questionnaire (Q2) was constructed as a contextualized exploratory measure for assessing participants' perceived aesthetic and cultural experience in embodied interaction with intangible cultural heritage (ICH). The questionnaire consists of 22 items across four dimensions: Emotional Resonance, Cognitive Meaning Construction, Action-based Meaning Perception, and Cultural Meaning Transmission. All items were rated on a 7-point Likert scale. The internal consistency coefficients reported below were estimated using an independent pilot sample ($n = 26$).

Dimension	Item	Item wording
Emotional Resonance ($\alpha = .789$)	ER1	During the experience, I felt genuinely emotionally moved.
	ER2	This installation evoked a deep emotional response within me.
	ER3	During my interaction with the installation, my emotional state changed in a meaningful way.
	ER4	After the experience ended, I could still feel a lingering emotional impression.
	ER5	During the experience, I felt an emotional resonance between myself and the atmosphere of the work.
	ER6	This work gave me a sense of inner tranquility and calmness.
Cognitive Meaning Construction ($\alpha = .752$)	CM1	Through this experience, I developed an understanding of the work that went beyond its surface-level visual appearance.
	CM2	The experience naturally evoked associations with Chinese traditional cultural contexts or my personal memories.
	CM3	Through the interaction, I perceived deeper meanings that the work sought to convey.
	CM4	This experience prompted me to actively interpret and reflect on the cultural meanings underlying the work.
	CM5	This work left me with a strong sense of meaning and gave me something to reflect on.
Action-based Meaning Perception ($\alpha = .868$)	AM1	The way I interacted with the work helped me perceive its rhythm and experiential qualities.
	AM2	My actions during the interaction contributed to how I experienced the work.
	AM3	The interaction process helped me understand aspects of the work beyond its visual appearance.
	AM4	The relationship between my actions and the system's feedback influenced how I understood the meaning of the work.

Dimension	Item	Item wording
Cultural Meaning Transmission (α = .826)	CT1	This work effectively conveyed culturally meaningful aesthetic qualities
	CT2	The work conveyed an aesthetic quality of simplicity, restraint, and balanced use of visual space.
	CT3	The work conveyed a natural and handcrafted quality associated with traditional craft practices.
	CT4	The work conveyed a sense of vitality through qualities of growth and movement.
	CT5	The work conveyed a sense of time through gradual change and transformation.
	CT6	The aesthetic qualities I perceived helped me understand the cultural meanings underlying the work.
	CT7	The interaction helped connect the aesthetic qualities of the work with their cultural context.